Milestone 1 Report

Studying uGLAD: A Deep Learning Approach to Conditional Independence Graph Recovery

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Group: g12_th_conditional-ind-graphs_ind | Track: Theoretical AI (Individual)
Student: Het Patel, AU2440266

1. Problem Statement

The problem addressed by our base paper, uGLAD [1], is recovering the structure of a conditional independence (CI) graph directly from data. Given samples of a set of variables, the goal is to find which variables stay connected once the effect of every other variable is removed. For Gaussian data this is the same as estimating a sparse precision matrix, since a zero entry in the precision matrix means the matching two variables are conditionally independent given the rest.

Classical methods pose this as one convex optimization problem, called the graphical lasso [3], and solve it with a single fixed regularization value that controls how sparse the recovered graph is. GLAD [2], the direct predecessor of uGLAD, replaced the classical solver with a small neural network that learns this regularization instead of fixing it, but GLAD needed the true graph during training to do so. uGLAD removes that requirement. It learns the same kind of network without ever seeing a ground truth graph, using only the input data.

Motivation. CI graphs are useful whenever the relationships between variables in a domain are not already known, for example gene regulation, healthcare data, or any setting with many variables and comparatively few samples. The uGLAD paper itself demonstrates this on gene regulatory network data, an anaerobic digestion dataset, and an infant mortality dataset.

Significance. If a learned optimizer can recover CI graphs without supervision and without hand tuning a regularization value, that changes how this problem is normally approached, since classical solvers depend heavily on that tuning step.

Assumptions used by the paper:

- Data is generated, or reasonably approximated, by a multivariate Gaussian distribution.
- The true precision matrix is sparse.
- Samples are independent and identically distributed.
- The number of variables D can be larger than the number of samples M.

Scope. At this stage our scope is to understand the problem uGLAD solves, read the paper closely along with its direct predecessor GLAD [2] and its two follow up papers, Neural Graphical Models [4] and Neural Graph Revealers [5], and study the survey by the same authors [6] to see how this problem is normally approached. This reading is still in progress. Our own specific research question, together with the exact baseline comparison and experimental design, will be finalized once this reading is complete, in time for the next milestone.

2. Review of Existing Work

We reviewed the papers closest to the problem uGLAD addresses, all from the same research line. The table below compares the problem each paper addresses, its method, its key assumptions, the metric it reports, and its limitation with respect to our reading.

Paper	Problem Addressed	Method	Key Assumptions	Metric	Limitation
uGLAD (Shrivastava, Chajewska, Abraham, Chen, NeurIPS 2022 Workshop) [1]	Unsupervised recovery of sparse Gaussian graphical models.	Unrolls the Alternating Minimization algorithm into a small neural network (GLADcell) that learns the regularization.	Gaussian data, sparse precision matrix, i.i.d. samples.	AUC, AUPR	Workshop paper, not a main track publication. No formal convergence proof beyond the underlying AM algorithm.
GLAD (Shrivastava et al., ICLR 2020) [2]	Sparse graph recovery via a learned optimizer.	Same unrolled architecture as uGLAD, trained with supervision.	Ground truth precision matrices available during training.	NMSE, probability of success	Needs labeled graphs, rarely available outside synthetic benchmarks.
Graphical lasso (Friedman, Hastie, Tibshirani, 2008) [3]	Classical sparse precision matrix estimation.	Block Coordinate Descent solving the l1-regularized maximum likelihood objective.	One fixed regularization value for the whole matrix.	Log-likelihood, recovery accuracy	Regularization value must be tuned separately, which is slow.
Neural Graphical Models (Shrivastava & Chajewska, ECSQARU 2023) [4]	Representing dependency structure over mixed variable types with proper inference.	Neural-network based multi-task learning framework over a given graph.	Allows general functional relationships, not only Gaussian.	Likelihood and query-based evaluation	Broader scope than plain CI graph recovery, adds model complexity.
Neural Graph Revealers (Shrivastava & Chajewska, ICML 2023 Workshop) [5]	Merging graph recovery with downstream probabilistic queries.	Extends the uGLAD-style architecture to support queries after recovery.	Builds directly on a recovered graph structure.	Evaluated on healthcare data tasks	Domain specific validation, less focus on CI graph accuracy alone.
Survey (Shrivastava & Chajewska, 2022) [6]	Summarize methods for recovering CI graphs.	Literature survey.	Not applicable.	Not applicable, review paper	Does not itself propose or test a method.

3. State-of-the-Art (SOTA) Position

The strongest and most directly relevant recent work from a major research lab is Causal Order, from Microsoft Research (Vashishtha et al., ICLR 2025) [7]. It studies how to recover graph structure when the only outside help available is an imperfect expert, such as a large language model, and shows that asking the expert for a variable ordering is more reliable than asking it directly for edges. It evaluates recovery quality using SHD, edge precision, edge recall and F1, which is the same evaluation language used across the CI graph recovery literature, including uGLAD's own follow up work.

A second, related Microsoft Research paper is Cond-FiP (Mahajan et al., TMLR 2025) [8], which learns to infer causal mechanisms across many datasets at once, instead of solving each dataset separately from scratch. This is close in spirit to how uGLAD learns a general optimizer instead of solving each precision matrix estimation problem independently, even though Cond-FiP targets directed causal graphs rather than undirected CI graphs.

Both papers work on directed causal graphs, while our base paper, uGLAD, works on undirected CI graphs. Our current position is that Causal Order is the closer SOTA anchor, since it shares uGLAD's evaluation language and its underlying concern with how far a learned or assisted method can be trusted without ground truth. This position may be refined once our reading of uGLAD's own follow up work is complete.

4. Representation

uGLAD represents the problem using an undirected Gaussian graphical model, also called a pairwise Markov random field. The nodes are the variables X1 through XD, modeled jointly as multivariate Gaussian. An edge is present between two nodes only when the corresponding entry of the precision matrix Θ is not zero, equivalently when the partial correlation between the two variables is not zero.

This representation fits the problem well for two reasons. It gives directly interpretable, direct-dependency information that plain covariance cannot give, since covariance alone cannot separate a direct relationship from one caused indirectly through other variables. It also keeps inference cheap, as explained in the Inference section below, because a jointly Gaussian model has closed-form marginals and conditionals.

5. Learning

In this Gaussian setting, structure and parameters are not cleanly separate. The nonzero pattern of Θ is the structure, and the values of Θ are the parameters, and both are recovered together as one matrix. Classical learning solves the l1-regularized maximum likelihood objective with a convex solver such as Block Coordinate Descent, using one fixed penalty chosen ahead of time.

uGLAD keeps the same objective but unrolls the optimization steps into a small recurrent neural network whose few parameters, the regularization values ρ and λ at each step, are learned by gradient descent on an unsupervised log-likelihood loss. No ground truth structure is needed during this training, so uGLAD performs unsupervised, score-based structure learning with a learned optimizer rather than a fixed one.

6. Inference

Once Θ is recovered, several useful queries become closed form, because the underlying model is jointly Gaussian. The marginal distribution of any subset of variables, and the conditional distribution of one variable given the rest, are both obtained directly from Θ. The partial correlation query, how strongly and in what direction Xi depends on Xj holding everything else fixed, is exactly the quantity uGLAD itself reports.

This is a key reason the Gaussian graphical model representation is attractive. In general graphical models, marginal and MAP inference are NP-hard in the worst case and need approximate methods such as belief

propagation, MCMC, or variational inference. Here, because the representation is Gaussian, exact inference reduces to matrix algebra on Θ.

7. Candidate Baselines and Metrics

The literature around uGLAD points to a natural baseline and metric set for this class of problem. Graphical lasso solved by Block Coordinate Descent is the classical baseline used inside the uGLAD paper itself, since it optimizes the exact same objective without learning. This course also lists constraint and score based structure learning algorithms, such as PC, GES and Hill Climbing, as standard Theoretical AI baselines; these would need to be adapted to undirected skeleton recovery for a fair comparison, since they are usually built for directed structure learning.

The metrics used consistently across this literature are Structural Hamming Distance (SHD), edge precision, edge recall, F1 score, AUC and AUPR for structure recovery quality, alongside held-out log-likelihood for how well the recovered model fits unseen data. These are the metrics we expect to use once our baseline comparison is finalized.

8. Timeline

Weeks	**Planned Work**
Weeks 1-2 (through M1)	Define the problem as posed by uGLAD, review the directly related papers, and pick Causal Order as the current SOTA anchor.
Weeks 3-4	Finish reading GLAD, Neural Graphical Models, Neural Graph Revealers and the survey. Reproduce uGLAD's public code.
Weeks 5-6 (to M2)	Finalize our specific research question and build the baseline pipeline (graphical lasso, PC, GES, Hill Climbing).
Weeks 7-8	Run initial experiments on synthetic Gaussian data and compare against the chosen baselines.
Weeks 9-10 (to M3)	Extend experiments to noise and missing data. Begin statistical analysis across multiple seeds.
Weeks 11-12 (to M4)	Consolidate results, revisit SOTA positioning, and write the final analysis.

References

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[7] A. Vashishtha, A. G. Reddy, A. Kumar, S. Bachu, V. Balasubramanian, and A. Sharma, “Causal Order: The Key to Leveraging Imperfect Experts in Causal Inference,” in International Conference on Learning Representations (ICLR), 2025.

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